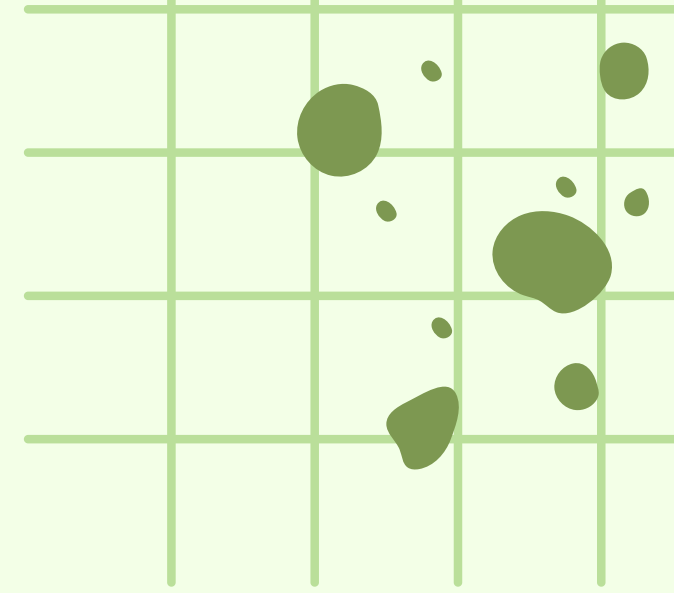
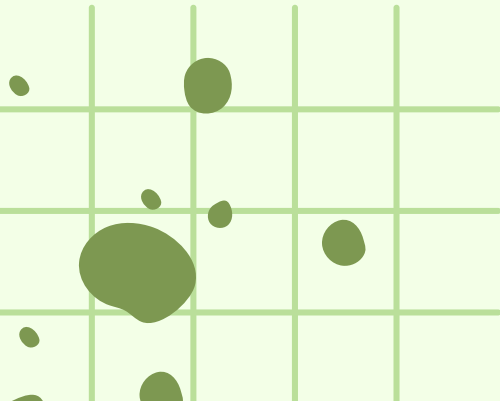




植物小小守護者：

對抗環境壓力

GEMS Taiwan 2026



LET'S SEE HOW MUCH YOU KNOW!

讓我們看看你知道多少！

Choose game mode - Kahoot!





QUESTIONS OF THE DAY

今天提問

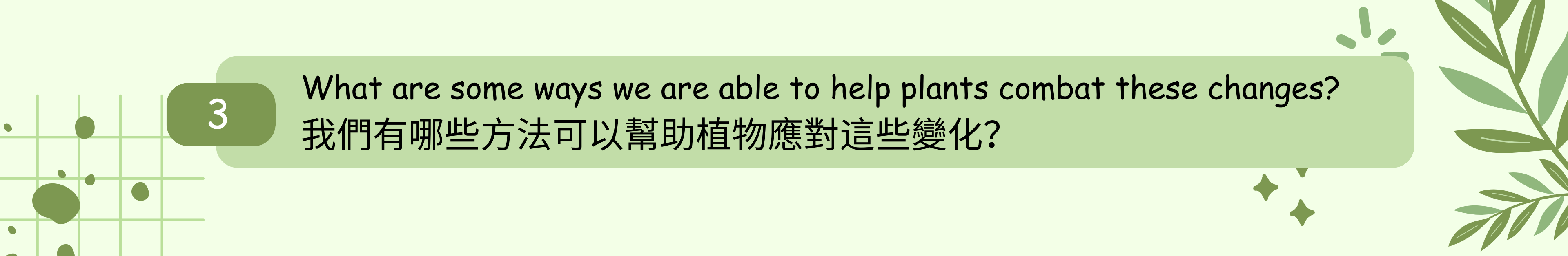
1

What happens when global warming gets worse and worse?
當全球暖化越來越嚴重時會發生什麼？

2

How might changes in temperature affect the way plants grow ?
溫度變化會如何影響植物的生長？

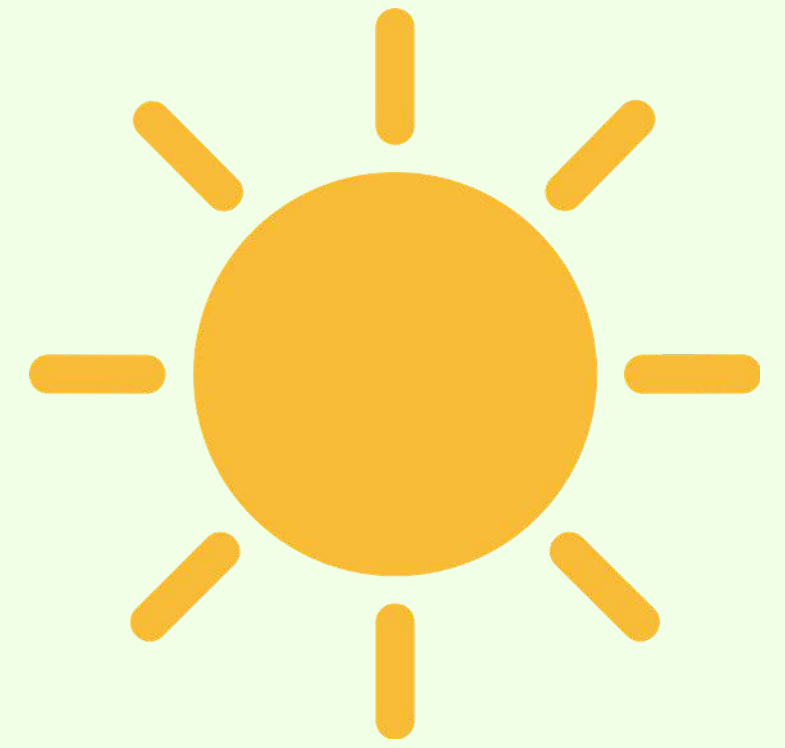
3



What are some ways we are able to help plants combat these changes?
我們有哪些方法可以幫助植物應對這些變化？

PROBLEMS IN TODAY'S WORLD

當今世界面臨的問題

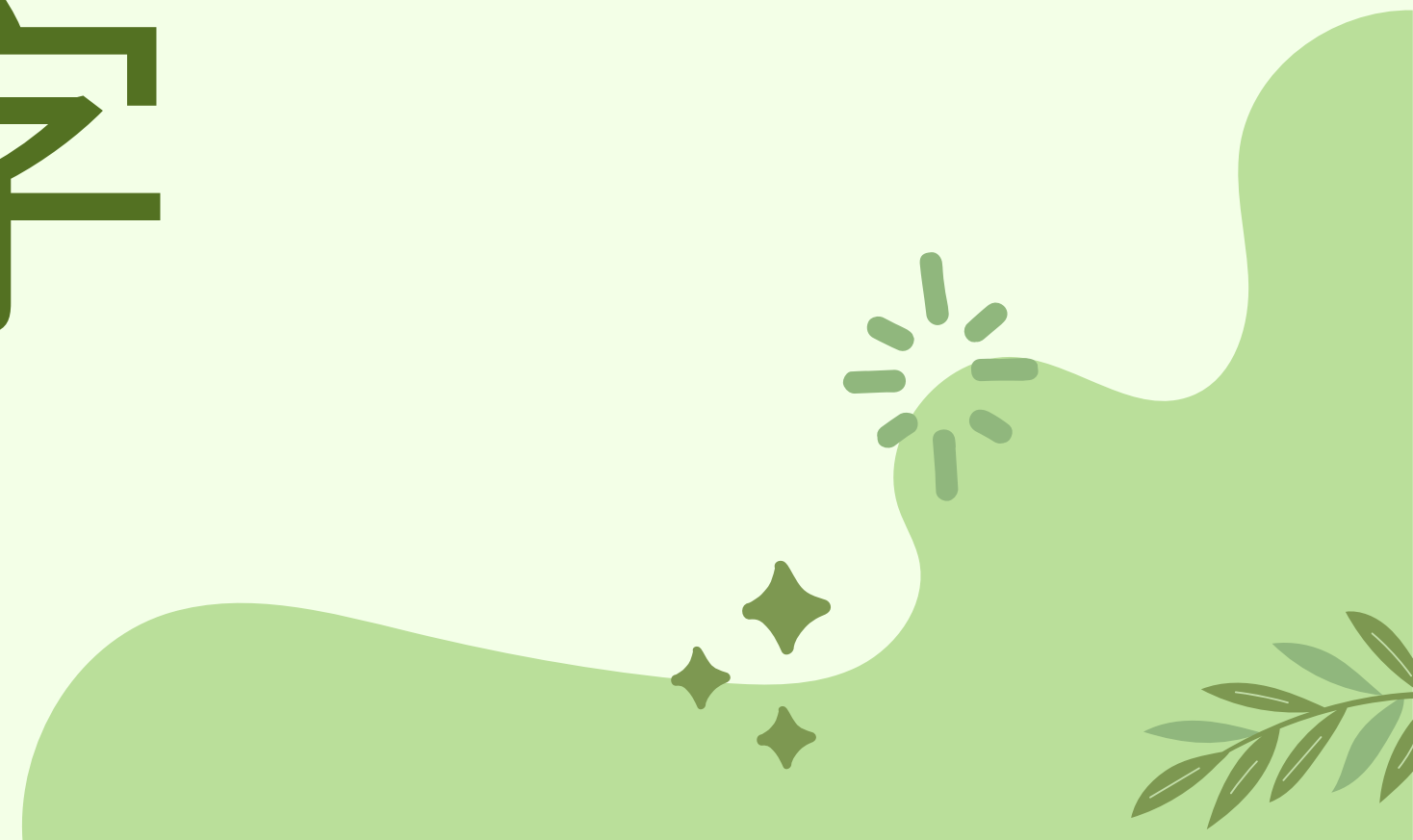


- Climate change causes the weather on Earth to rise
氣候變遷導致地球的天氣變得更熱
- Many of the crop we eat will have trouble growing well
我們吃的許多農作物將難以正常生長
- People can't find enough food to eat
人們可能無法找到足夠的食物來吃。



01 Vocabulary

單字



Climate Change

氣候變遷

- Long-term changes in Earth's temperature and weather patterns, often caused by human activity.

地球長期的氣溫與天氣變化，通常由人類活動造成。

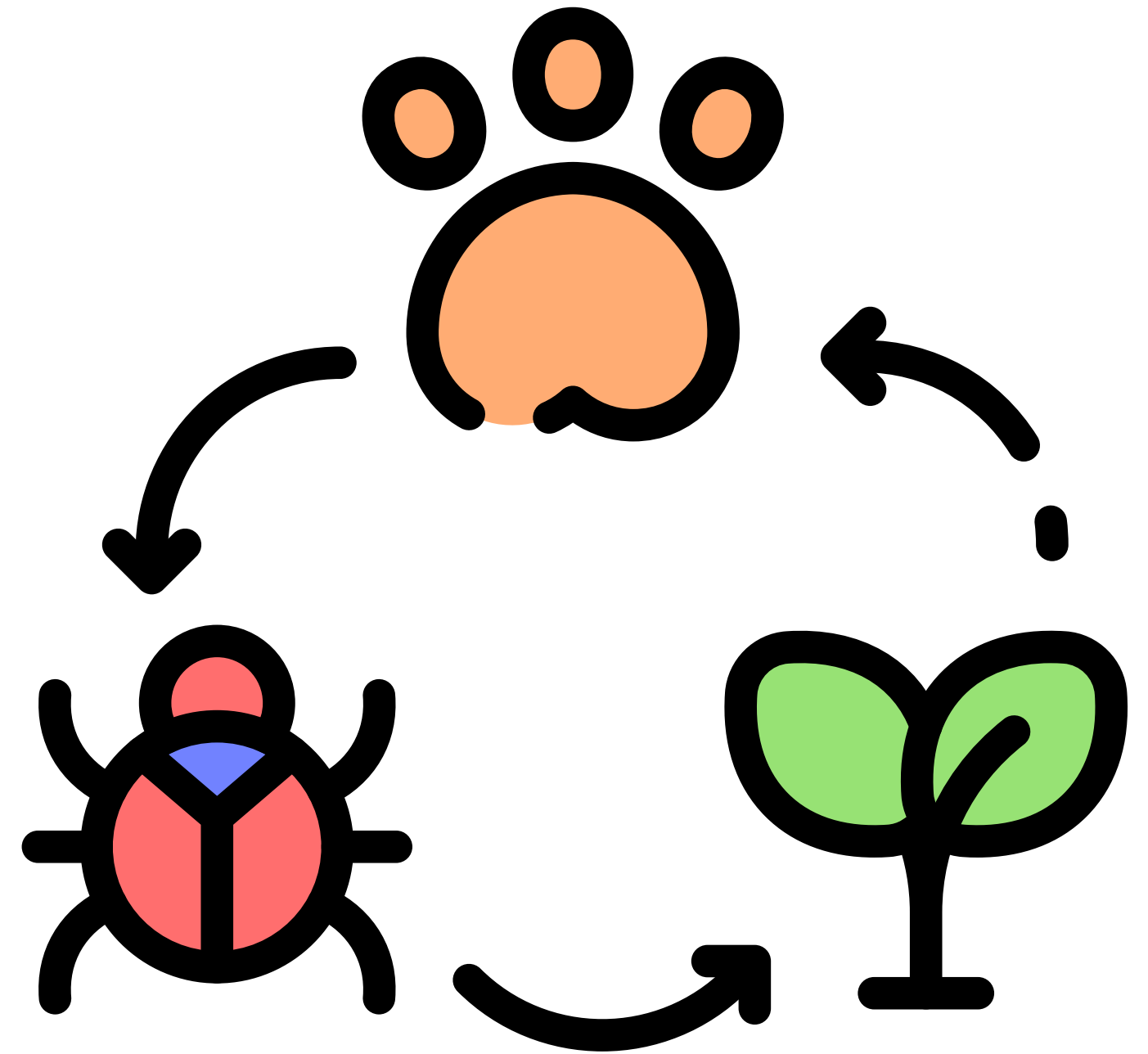
Example 例子：Global warming is part of climate change.

全球暖化就是其中一種氣候變遷



Biotic 生物性

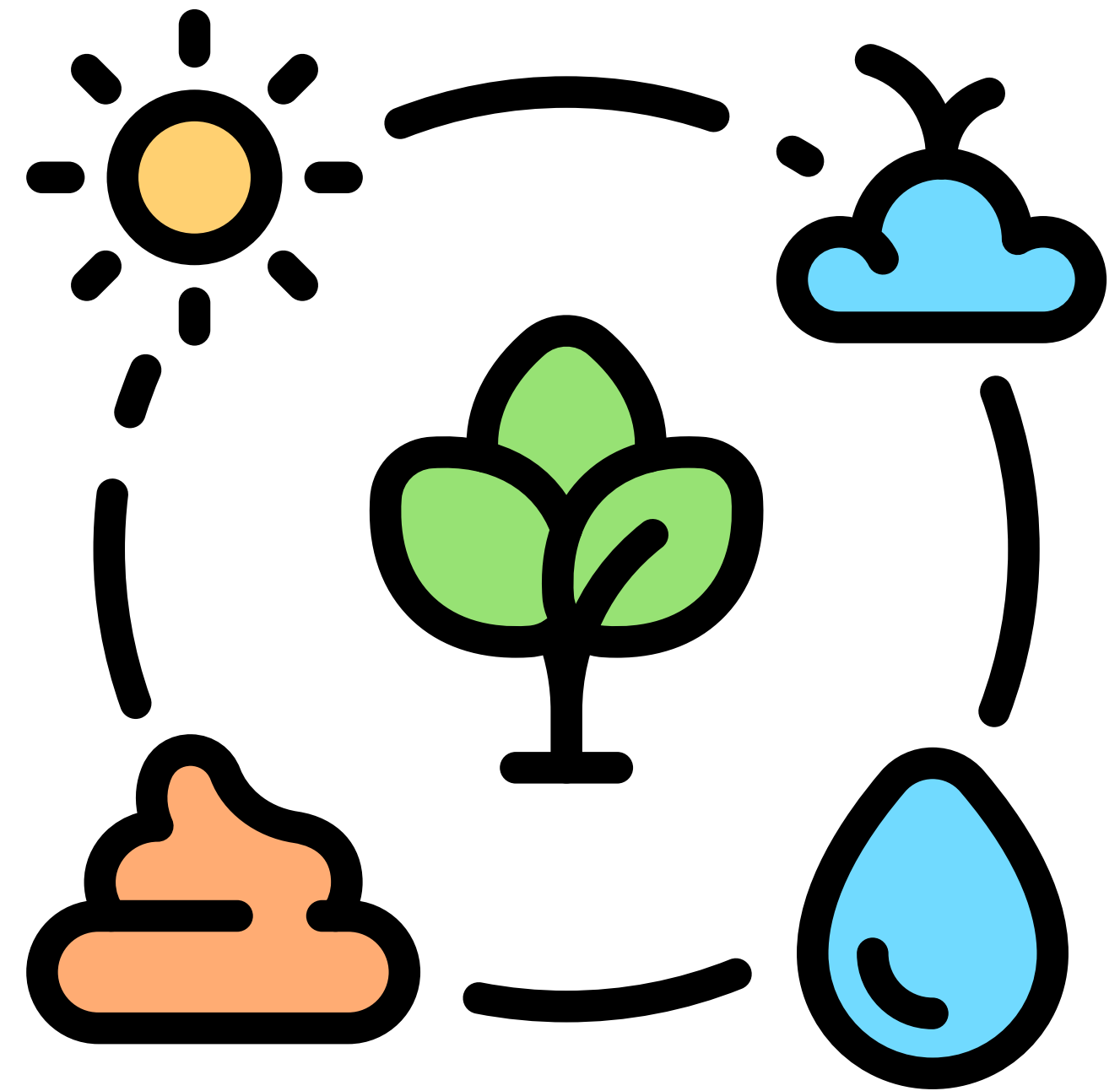
Living parts of an environment,
like plants, animals, and bacteria.
環境中所有的生物，例如植
物、動物和細菌。



Abiotic 非生物性

Non-living parts of an environment, like sunlight, water, air, and temperature.

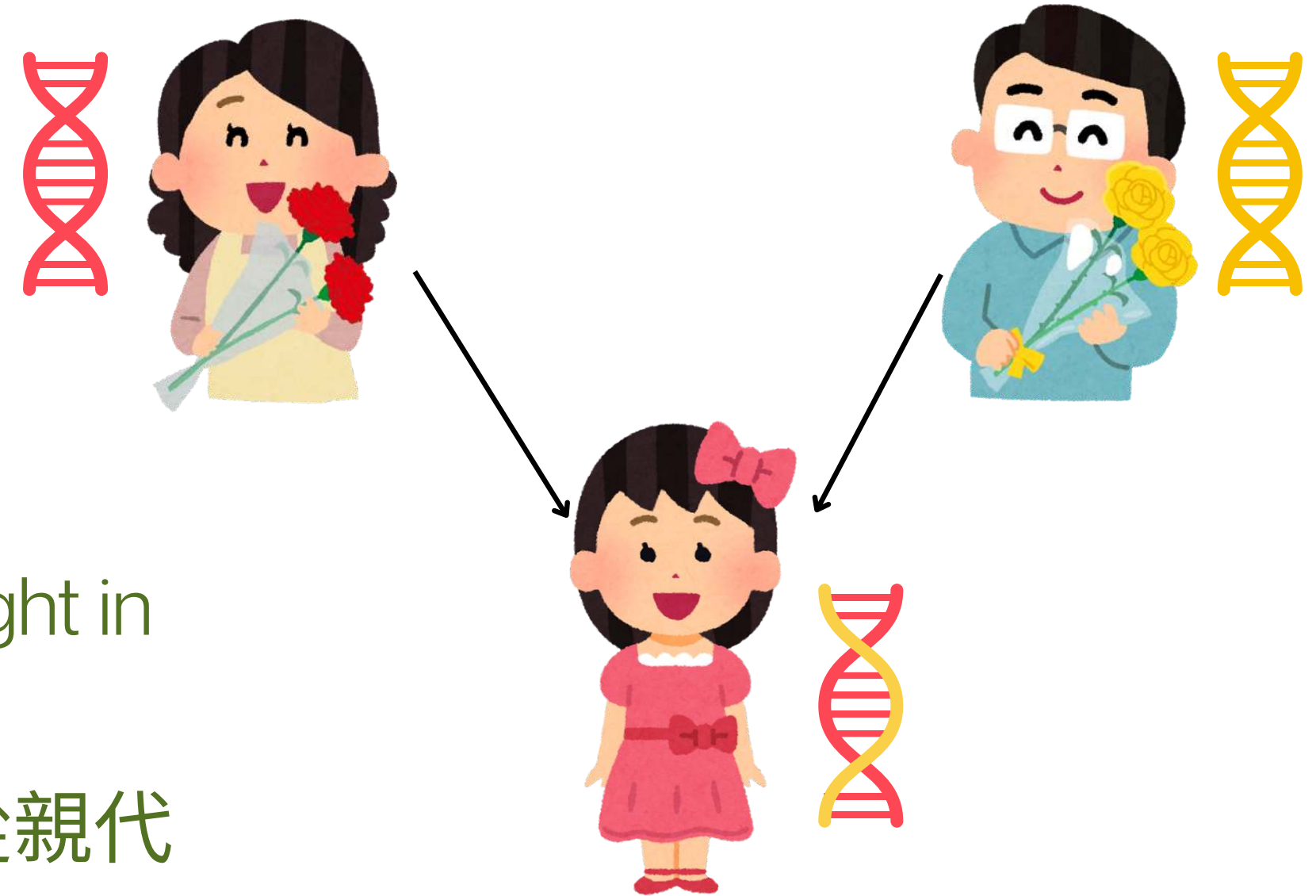
環境中的非生物因素，例如陽光、水、空氣和溫度。



Genetics

基因

- The study of how traits (like eye color or height in plants) are passed from parents to children.
- 研究特徵（如眼睛顏色或植物高度）如何從親代傳給後代的科學。

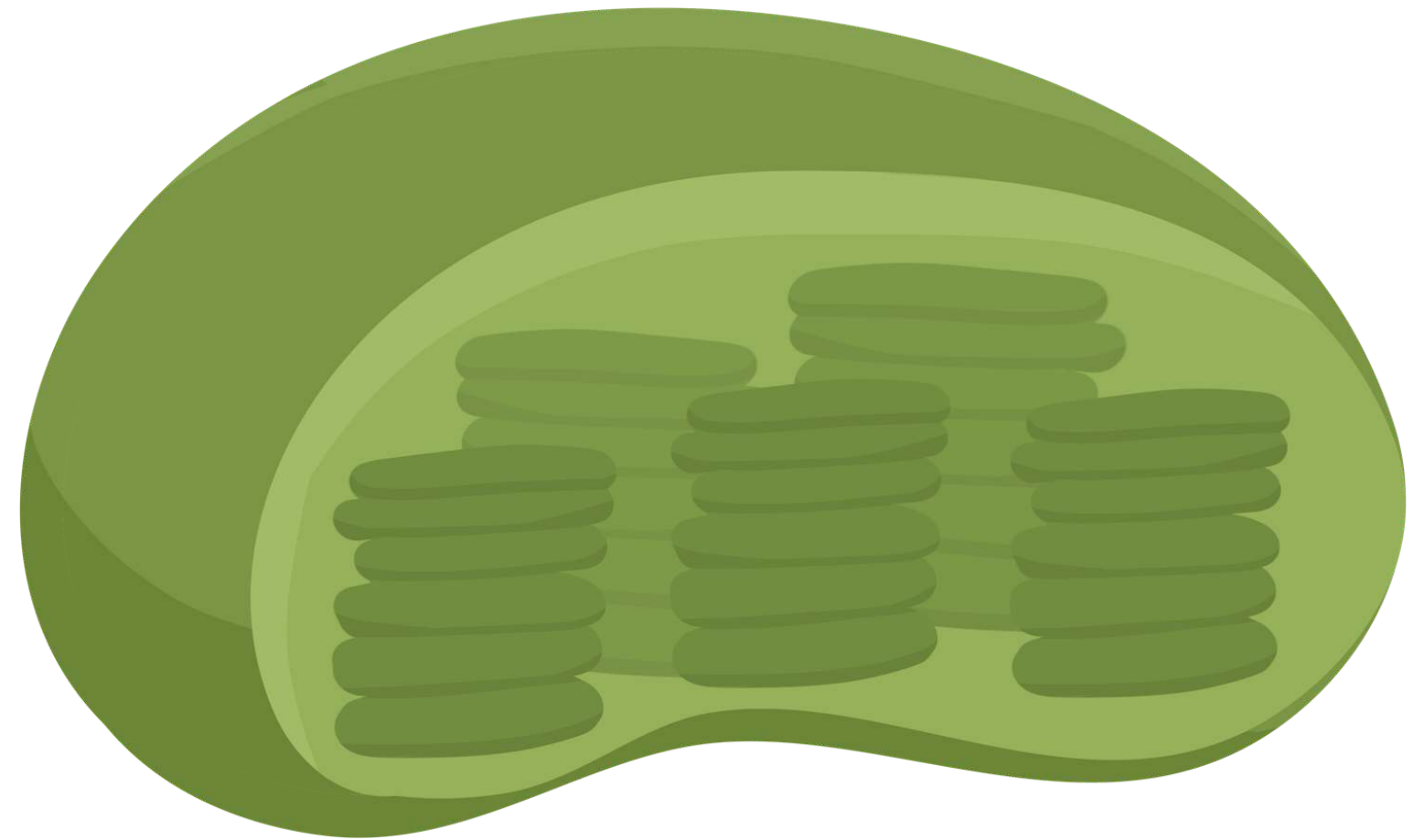


Chlorophyll

葉綠素

- A green pigment in plants that helps capture sunlight for photosynthesis

植物中的綠色色素，幫助吸收陽光進行光合作用。

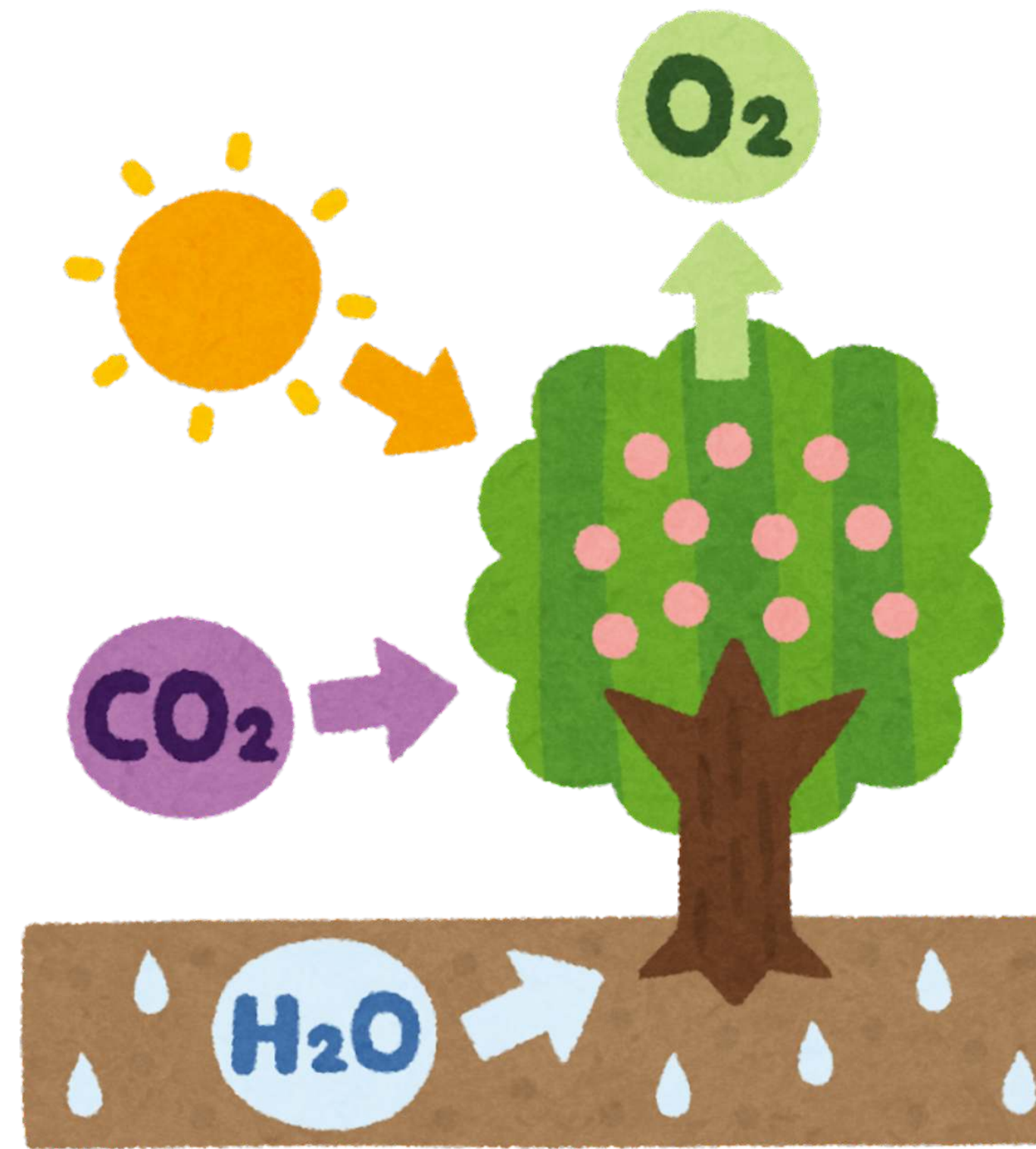


Photosynthesis

光合作用

The process where plants use sunlight, water, and carbon dioxide to make their own food (sugar).

植物利用陽光、水和二氧化碳製造自己食物（糖）的過程。



02

Photosynthesis

光合作用



Photosynthesis

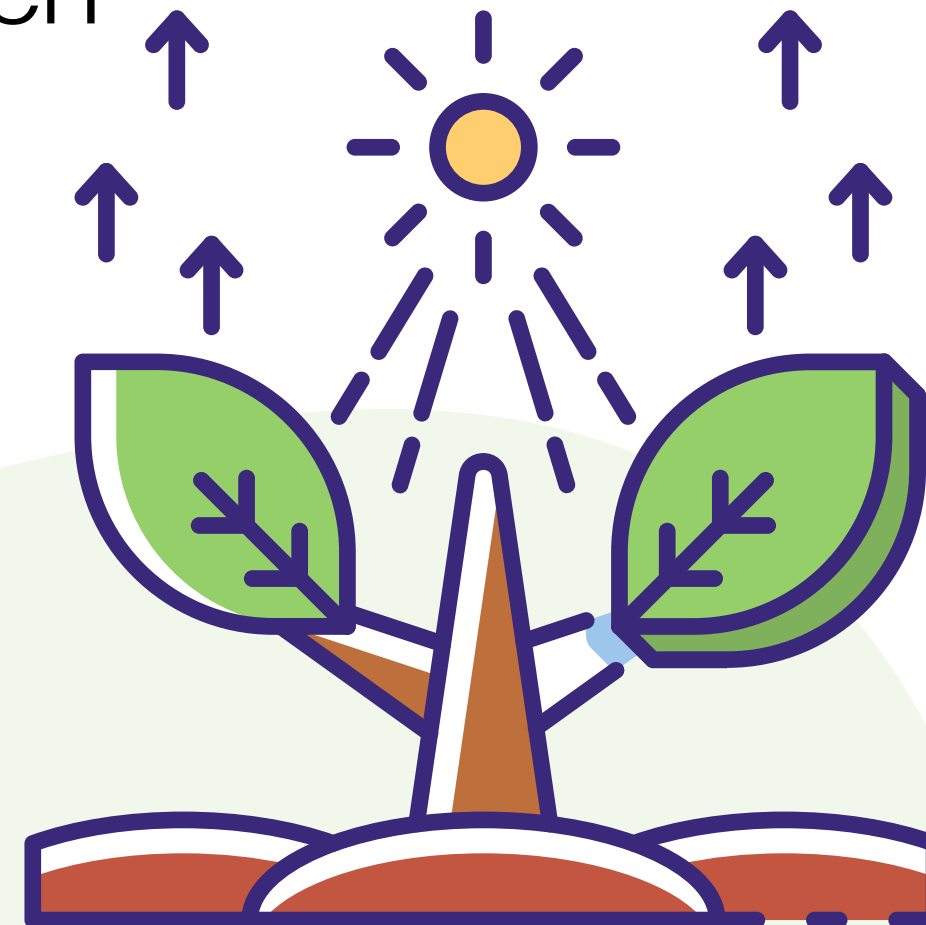
00:00 – 1:42



Photosynthesis 光合作用

Main Ideas 重要概念：

- $6\text{CO}_2 + 6\text{H}_2\text{O} + \text{Light Energy} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$.
二氧化碳 + 水 + 陽光 → 葡萄糖 + 氧氣
- Carbon Dioxide + Water + Sunlight $\rightarrow$ Glucose + Oxygen
- Photosynthesis occurs in the Chlorophyll
光合作用發生在葉綠體中



03

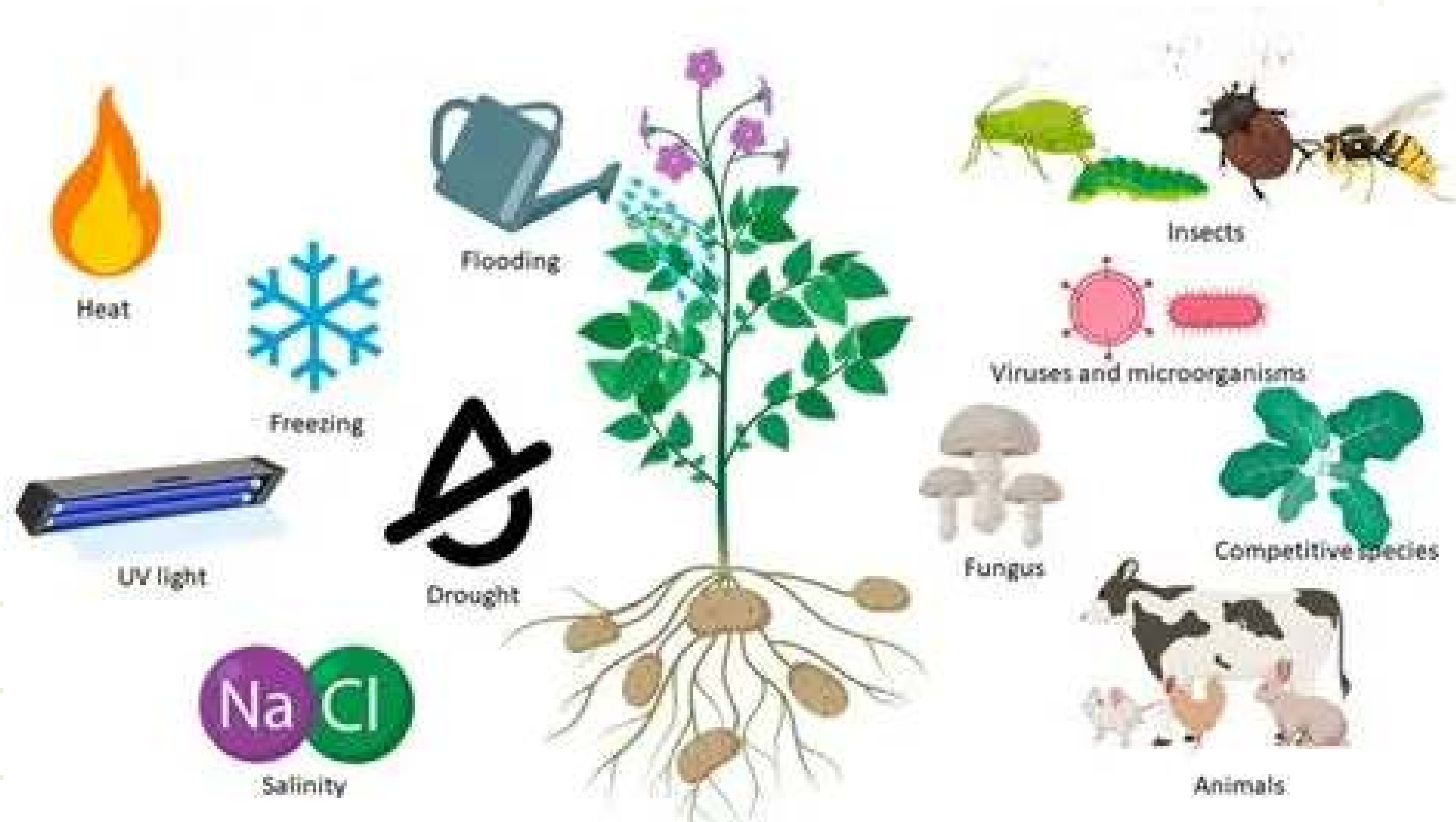
Plant Stress

植物壓力



不同種類的植物壓力

TYPES OF STRESSES



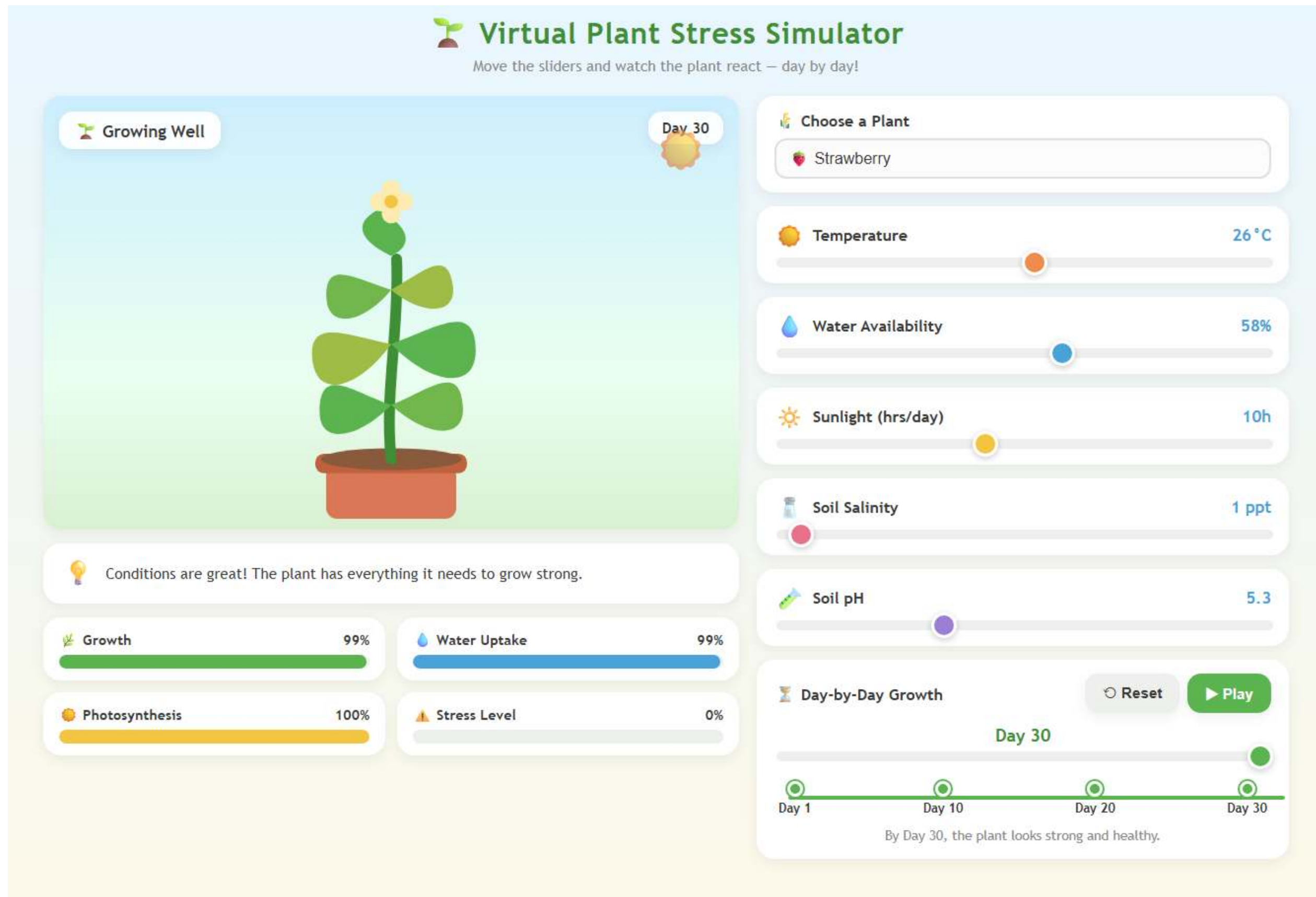
WHAT MIGHT HAPPEN TO A PLANT UNDER STRESS?

在壓力（逆境）下，植物可能會出現以下情況

Photosynthesis decreases 光合作用減少

- Growth slows down 生長變慢
- Leaves turn yellow or wilt 葉子變黃或枯萎
- Lacks water and dries out 缺水、容易乾枯
- Leaves fall off 葉子掉落
- Can't flower or produce fruit normally 無法正常開花或結果
- May die in severe conditions 嚴重時可能死亡

看看植物在受壓下會發生什麼事？



04

實驗：

植物受熱時會發生什麼事？

<https://www.youtube.com/watch?v=gCHW7f88bAc>

Photosynthesis Visualized

光合作用的視覺化呈現

Materials Needed所需材料：

- 1 clear cup
1 個透明杯子
- Water
水(室溫水和溫水)
- ¼ tablespoon of baking soda
1/4 湯匙小蘇打
- Dish soap
洗碗精
- Thin, soft Leaves (Spinach)
薄而柔軟的葉子 (菠菜)
- Syringe
注射器
- Plant growing Light
植物生長燈



STEP 1

- Prepare cup of water each with 100 mL of water of the correct temperature

準備數個杯子，並在每個杯子中加入 100 mL

符合實驗要求溫度的水

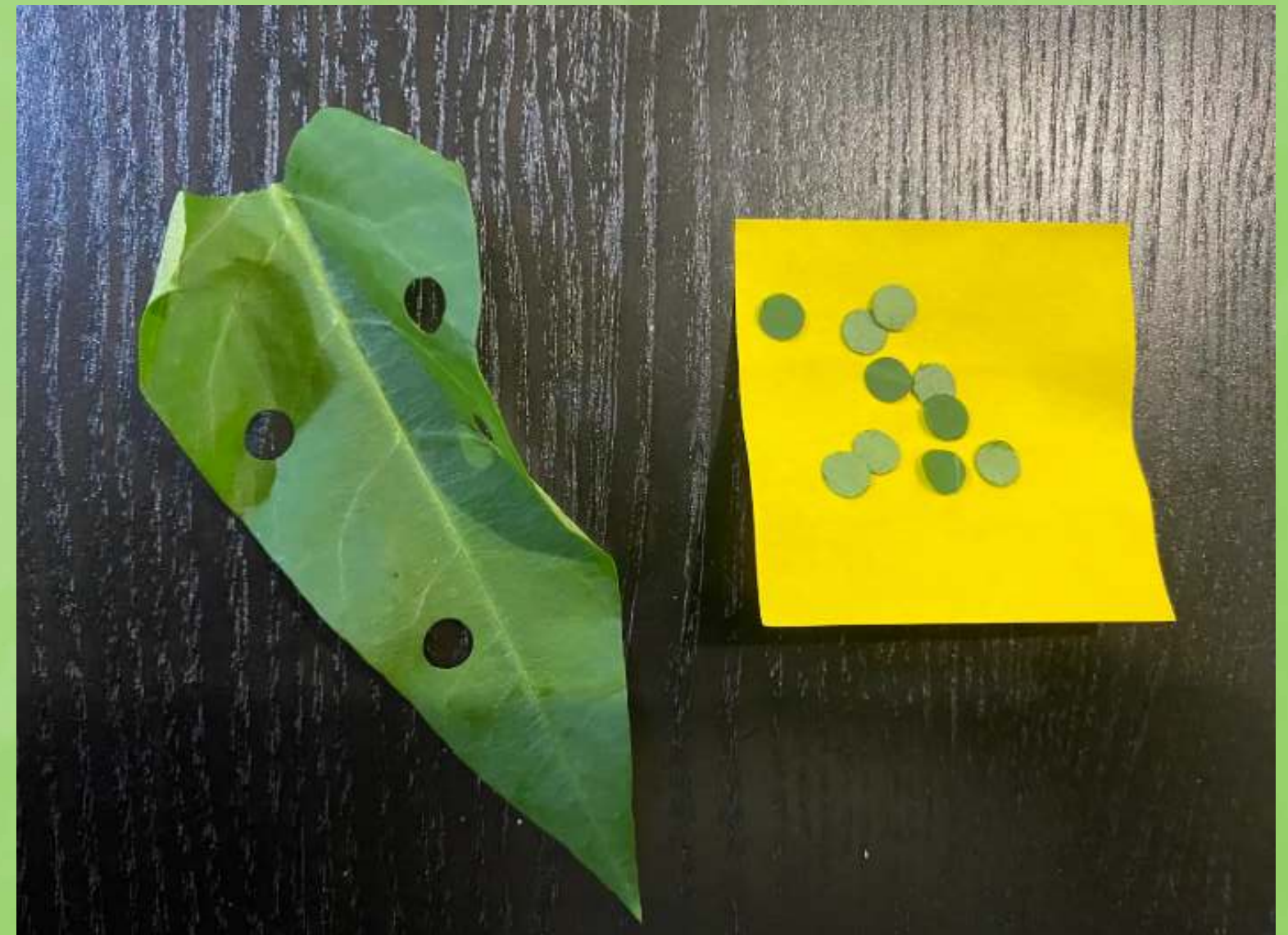
- In your teams beaker add $\frac{1}{4}$ tablespoon of baking soda and a small amount of dish soap and stir
- 在小組的杯中加入 $\frac{1}{4}$ 湯匙小蘇打粉 和少量洗碗精



STEP 2

- Using a hole-puncher or a pen cap, each group should cut out **5** disks of leaves

用打孔器剪出5片葉圓片



STEP 3

- Put **5 of the leaf disks** into the syringe and press down until there is barely any space
將**5片葉片**圓片放入注射器中，然後往下壓，直到幾乎沒有空隙。
- **Be careful not to damage the leaves!**
小心不要弄壞葉片！



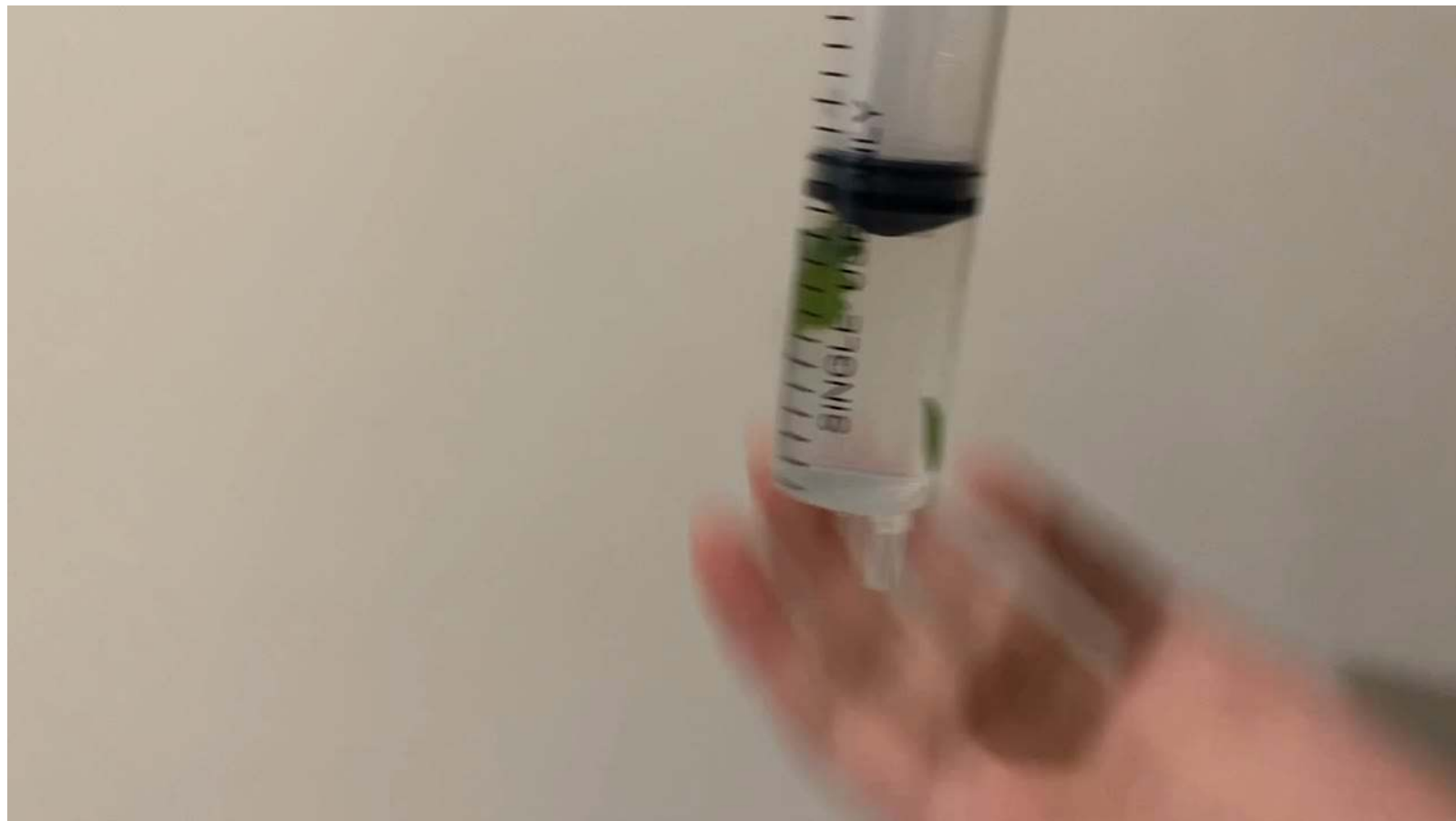
STEP 4

- Suck up **10 mL** of water into the syringe
在注射器中吸入**10毫升水**。
- You will see that the leaves are floating above the water
你會看到葉片漂浮在水面上方。



STEP 5

- Using your thumb, cover the hole at the end of the syringe
用拇指堵住注射器末端的開口。
- With your thumb still covering the hole, pull back the plunger and gently shake the syringe
在拇指仍然堵住開口的情況下，向後拉活塞，並輕輕搖晃注射器。



STEP 6

- Check to see if **all 5 of your leaf disks** have sunk to the bottom of the syringe
檢查5片葉片是否都已沉到注射器的底部。

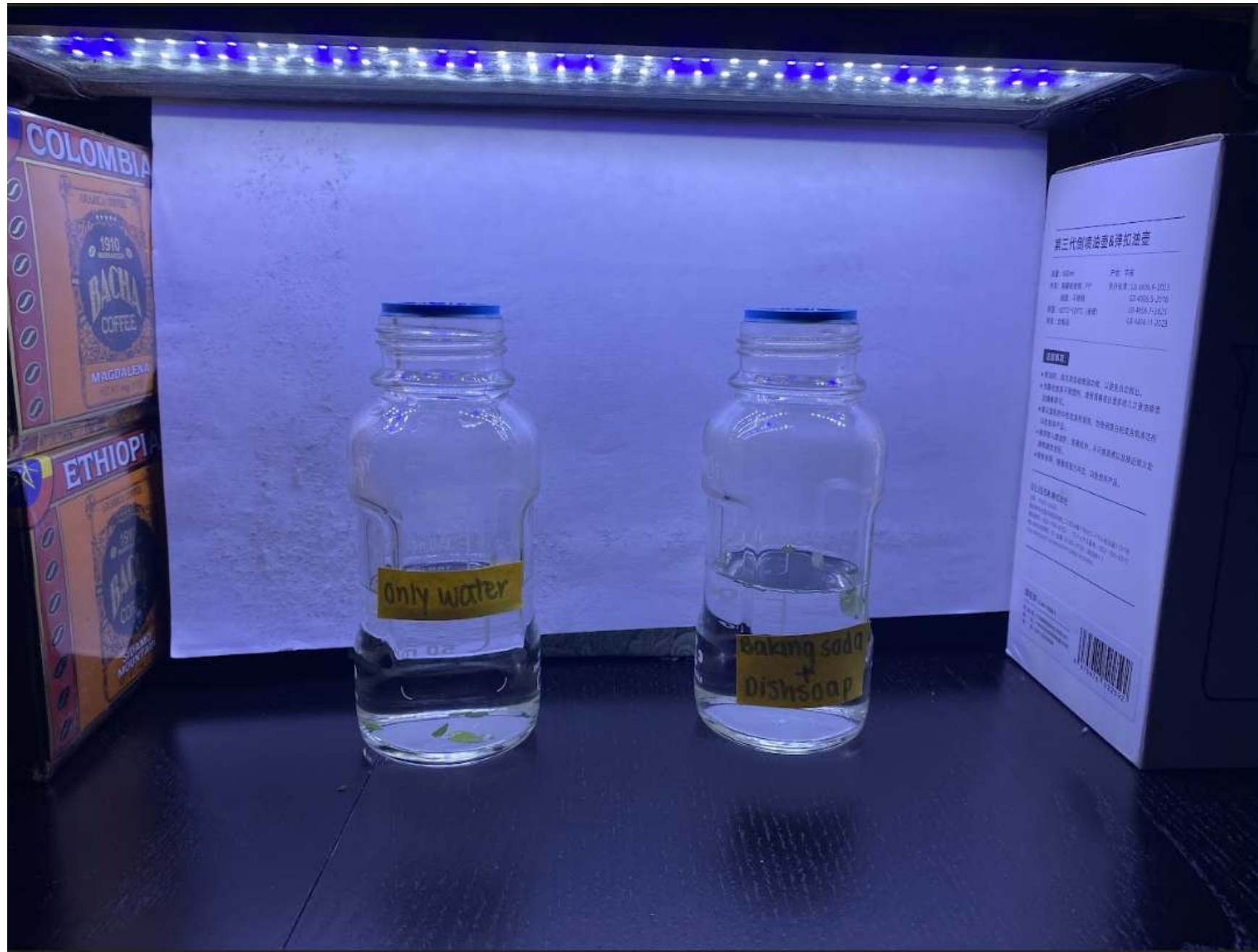


STEP 7

- Dump out all of the contents of the syringe into its according cup of water
將注射器中的所有內容倒入對應的水杯中。
- It should look like this:
應該會像這樣：



STEP 9



- Put a plant growing light over the 2 cups of water and observe what happens in the next few minutes
在兩個水杯上方放置植物生長燈，並觀察接下來幾分鐘內會發生什麼變化。

STEP 8

組別編號:

時間	浮起的葉片數量
2 分鐘 	
4 分鐘	
6 分鐘	
8 分鐘	
10 分鐘	
12 分鐘	
14 分鐘	
16 分鐘	
18 分鐘	
20 分鐘	

- Using a timer, record the time everytime a new leaf floats up on the peice of paper given to the group
使用計時器，每當有一片新的葉片浮到水面時，就記錄一次時間
- Work on the 2 questions below with your team
與你的團隊一起完成以下兩個問題。

Experiment Prediction

實驗預測

Hypothesis: What will happen | 假設：會發生什麼？

1

Discuss with your group and predict what will happen to the leaves
與你的小組討論並預測葉片會發生什麼變化

Each group sends one member to write their results on the board

每組派一人到前面把結果寫在板上

組別編號	水溫	葉片是否浮起來？有幾片？
組別 1	溫水	
組別 2	室溫	
組別 3	室溫	

Experiment debrief & Discussion

實驗回顧與討論

Discussion Question | 討論問題:

1

Why did the leaves in each temperature begin to float at different times?
為什麼在不同溫度下，每組葉片浮起的速度不同？

Explanation | 解釋

- Temperature affects how fast the leaf disks produce oxygen during photosynthesis.

溫度會影響葉片圓片在光合作用中產生氧氣的速度。

- At a moderate temperature, conditions are ideal, so oxygen is produced quickly and leaves float the fastest.

在適中的溫度下，條件最佳，所以氧氣產生最快，葉片最先浮起。

- At a high temperature, the leaf is under heat stress, so photosynthesis slows down and little or no oxygen is produced.

在較高溫下，葉片受到熱壓力，光合作用變慢，氧氣產生很少甚至沒有，葉片會浮不起來。

Experiment debrief & Discussion

實驗回顧與討論

Discussion Question | 討論問題:

2

How do the materials in the experiment affect how we can observe photosynthesis?
實驗中使用的材料如何影響我們觀察光合作用的方式?

Explanation | 解釋

- Baking soda provides carbon dioxide (CO_2), which is used in photosynthesis.
- 小蘇打提供二氧化碳 (CO_2)，這是光合作用所需的物質。
- Dish soap helps the CO_2 stick better to the leaf disks, making the reaction easier to observe.
- 洗碗精幫助二氧化碳更容易附著在葉片上，使反應更容易觀察。
- The syringe removes all air pockets from inside the leaves. That is why the leaf disks float at first but sink after the air is removed.
- 針筒會把葉片內的所有空氣排出，因此葉片一開始會漂浮，但當空氣被抽出後就會下沉。
- It shows how photosynthesis produces oxygen, which causes the leaf disks to float when light is present.
- 這個實驗顯示光合作用會產生氧氣，使葉片在有光照時再次浮起





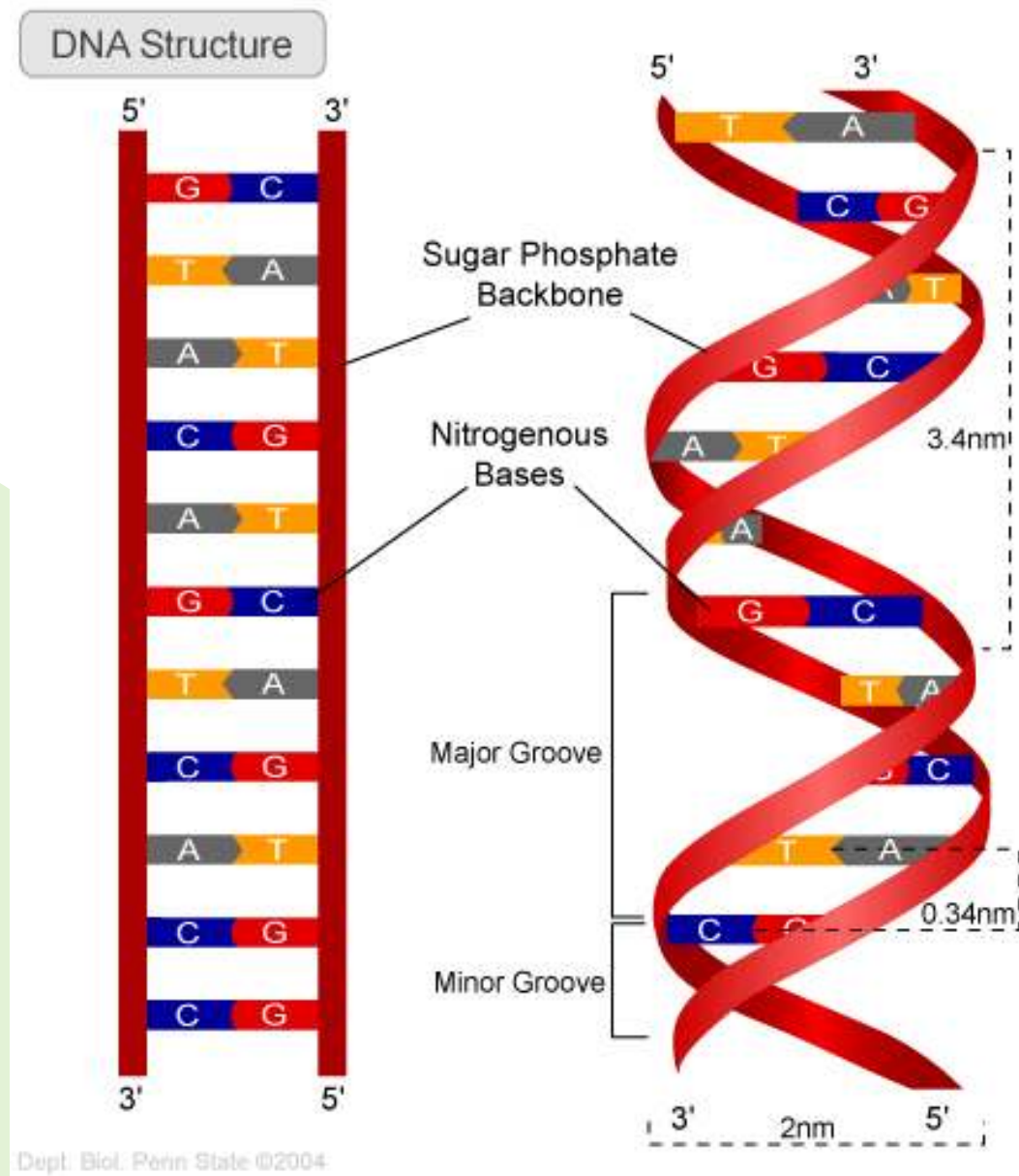
05

The Role of Science

如何用科學解決植物 面臨的問題？



DNA & GENETICS (基因)



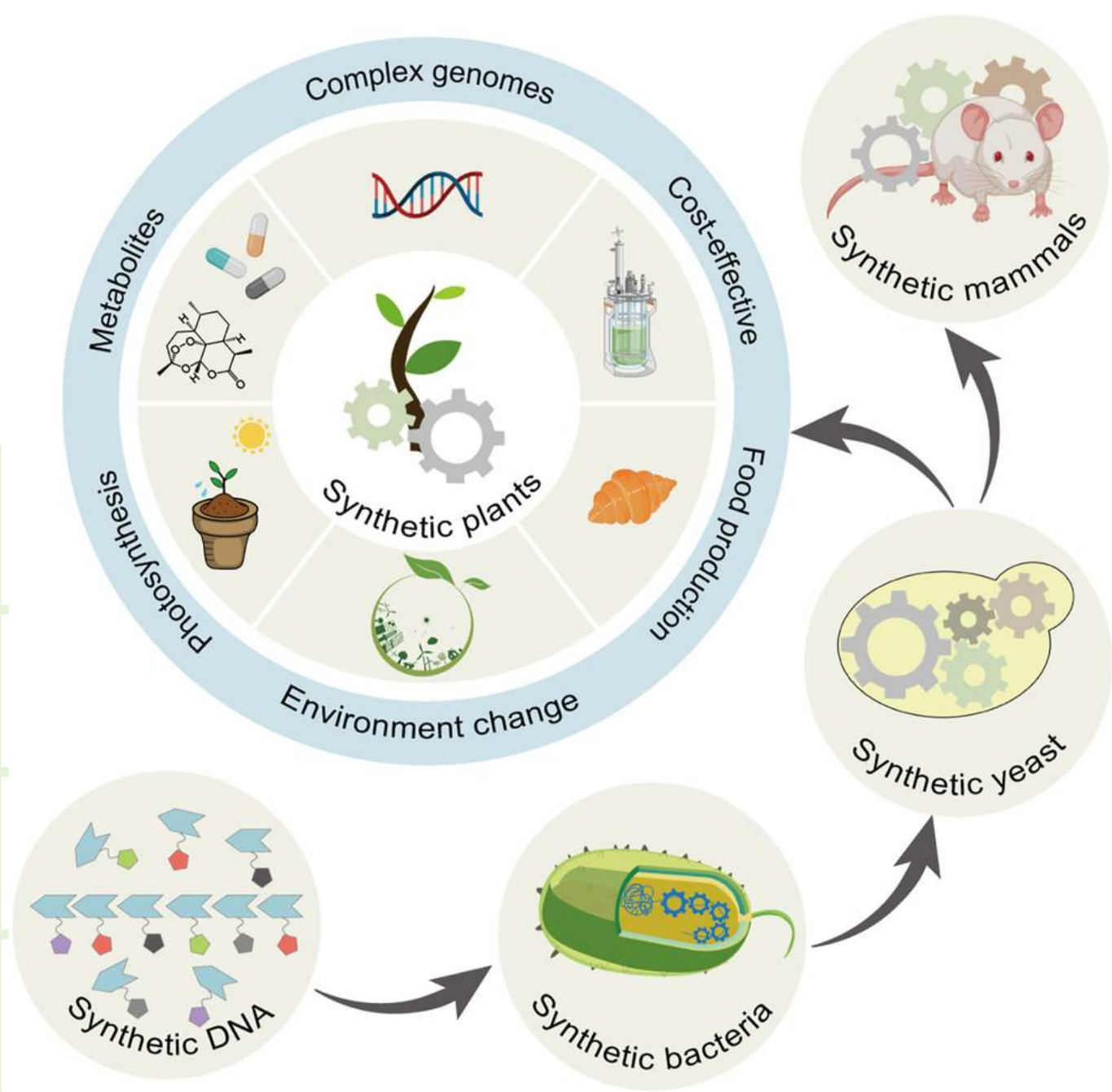
- DNA 存在於所有生物中，它會決定我們長怎樣，比如說眼睛、頭髮是什麼顏色。
- 它就像一組說明書
- 它告訴細胞如何生長以及要做什麼

DNA 是由四種「字母」組成的「代碼」：**A, T, C, G**
這些字母的排列順序提供指令，決定生物如何生長與運作。

A 配對 T； C配對G

WHAT IS SYNTHETIC BIOLOGY?

合成生物學是什麼？

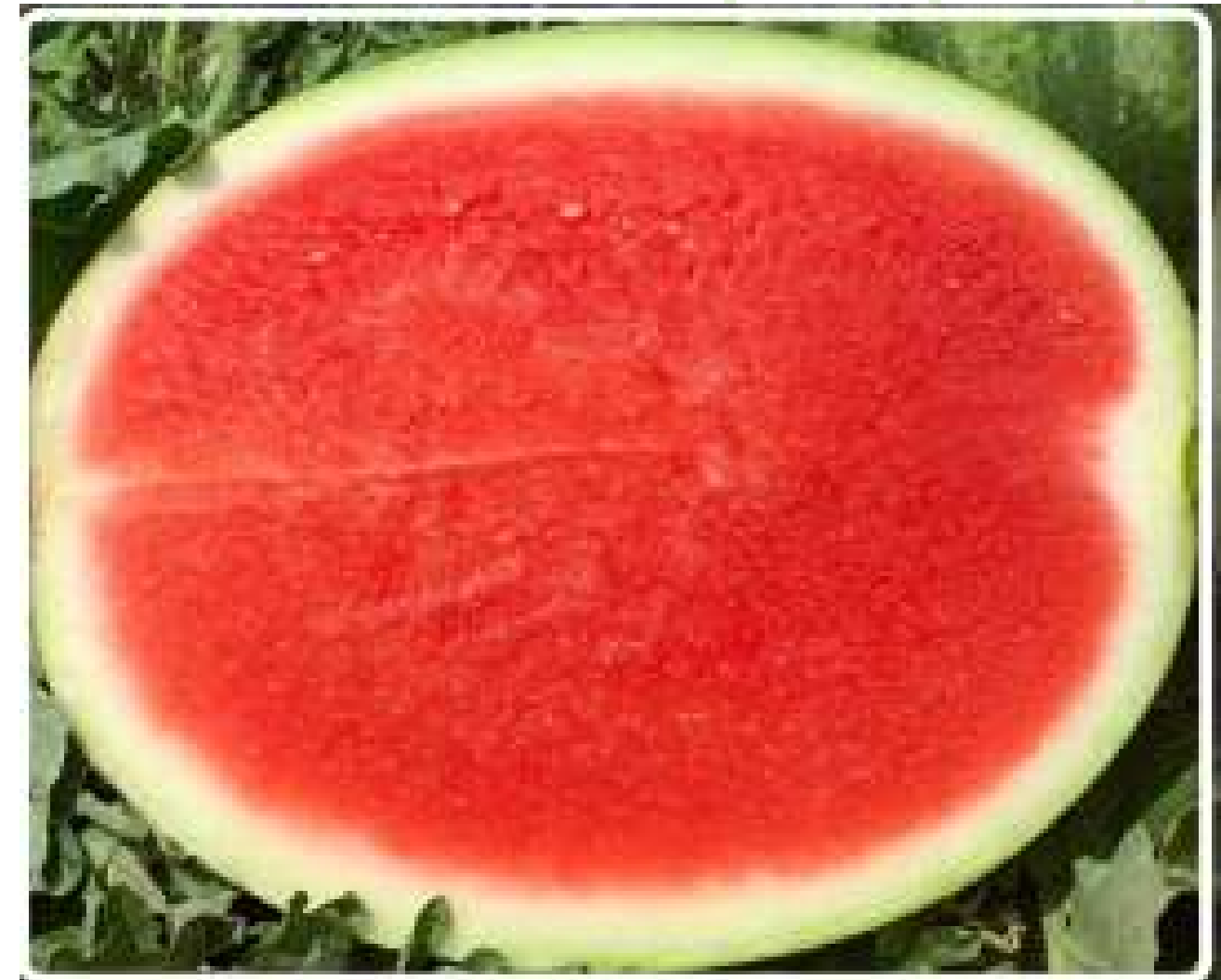


- Synthetic biology is when scientists change or design parts of a living things DNA
合成生物學是指科學家改變或設計生物體DNA的一部分
- They use cells like tiny building blocks
他們把細胞當作微小的積木來使用。
- Goal: help living things do specific useful jobs
目標：幫助生物完成特定有用的工作。
- Scientists can add or adjust instructions
科學家可以加入或調整這些指令
- This can change what a cell or organism can do
這樣可以改變細胞或生物體能做的事情

EXAMPLES OF SYNTHETIC BIOLOGY IN REAL LIFE

合成生物學在現實生活中的應用範例

- Fruits like Watermelon, Grapes, and oranges with no seed
像西瓜、葡萄和橙子這類無籽水果
- Food that doesn't spoil as fast
不容易變質的食物
- Fruits and Vegetables that won't be harmed by pests
不容易被害蟲破壞的水果和蔬菜
- Fruits that have a sweeter taste that more people like
口感更甜、更多人喜歡的水果



★ 我們在做什麼？(超級簡單版) ★

1. 目標 (GOAL)



幫植物平平安安長大
我們要像醫生一樣，讓植物健康！

2. 小小偵探和工廠 (TINY SENSORS & FACTORIES)



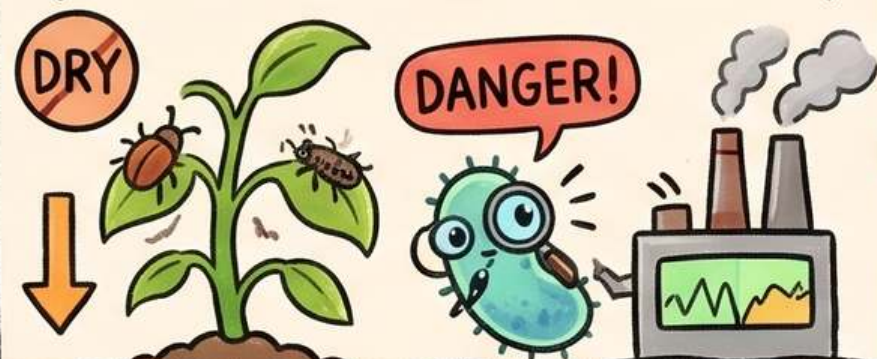
我們把小小的細菌變成超級英雄！

3. 給細菌指令 (GIVING INSTRUCTIONS)



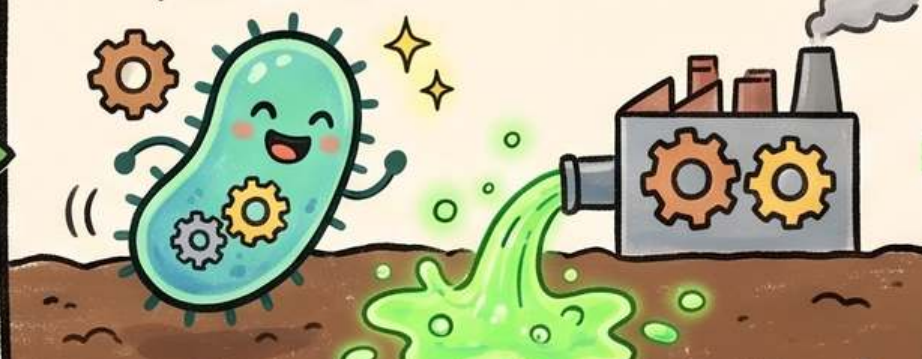
教細菌「什麼時候」和「怎麼做」

4. 當植物有危險 (WHEN PLANT IS IN DANGER)



偵植物有危險，賣打的工廠

5. 細菌變身！釋放魔法藥水 (BACTERIA TRANSFORM!)



釋放「植物藥水」來幫忙！

6. 成果 (RESULT)



植物擊退蟲蟲，也撐過惡劣天氣！

06

DNA Card Game

卡牌遊戲



DNA 抽鬼牌



- Each player gets a hand of cards.

每位玩家拿到一手卡牌。(5人組：8 張； 4人組：10張)

- Cards show DNA sequences and a plant.

卡牌上有DNA鹼基 (A、T、C、G) 和一種植物。

A 配對 T； C 配對 G

- Both Sequence and Plant needs to match in order for the pair to be thrown out. Take turn to draw the cards from each other in order to match the pairs of the cards

序列和植物都必須相同，才能把這一對牌丟出，然後輪流互相抽牌，嘗試配對卡牌。

- Mutant cards are the “Jokers” and contain a random letter, people taking the “Jokers” will lose.

突變卡是「鬼牌」，上面會有隨機的字母，最後擁有鬼牌的人會輸。

- First player to use all cards or complete plant sets wins.

第一個出完牌的玩家獲勝。



07

Fun Online Game

有趣的線上遊戲



Fun Online Game

有趣線上遊戲

🌱 iGEM SYNTHETIC BIOLOGY PROJECT

CROPSHIELD

BIOREACTOR DEFENDER

Our Mission: We've engineered a **sealed bioreactor** housing plant-growth-promoting rhizobacteria (PGPRs). When they sense environmental or biological stress, they release targeted protective compounds — safely, without releasing live bacteria into the environment.

You control the **bioreactor drone**. Select the right protectant and intercept each stress before it damages the crop. Survive each round and grow your plant to harvest!

← **A** → **D** Move drone

1 2 3 4 Switch protectant

Heat Protectant → Heat Stress

Antifungal → Fungal

SHIFT / Z Dash

E Rhizosphere Pulse

Saline Protectant → Salinity

Anti-Insect → Aphids & Caterpillars

▶ DEPLOY DRONE

08 Arts & Crafts

手作活動

Link: [Self-Watering Planter from a 1.5L Coke Bottle](#)
[#SelfWateringPlanter](#) [#PlasticBottleDIY](#) [#CokeBottleHack](#)



Self-Watering Plant Device



Materials Needed 所需材料：

- Clear PET bottle with cap
透明PET寶特瓶（含瓶蓋）
- Rope/ String
繩子／細繩
- Scissors
剪刀
- Soil
土壤
- Seed
種子
- Tape
膠帶

STEP 1

- Take your clear PET-bottle and draw a straight line across the bottle as shown in the picture below
拿出透明的PET寶特瓶，按照下圖在瓶身上畫一條直線



STEP 2

- Using cutter and. scissors carefully cut according to the line
用美工刀和剪刀沿著畫好的線小心剪開
- **If you need help, raise your hand and wait**
如果需要幫助，請舉手等待



STEP 3

- Next take your bottle cap and punch a hole big enough to pass a string through
接著拿起瓶蓋，在上面打出一個足夠讓繩子穿過的洞



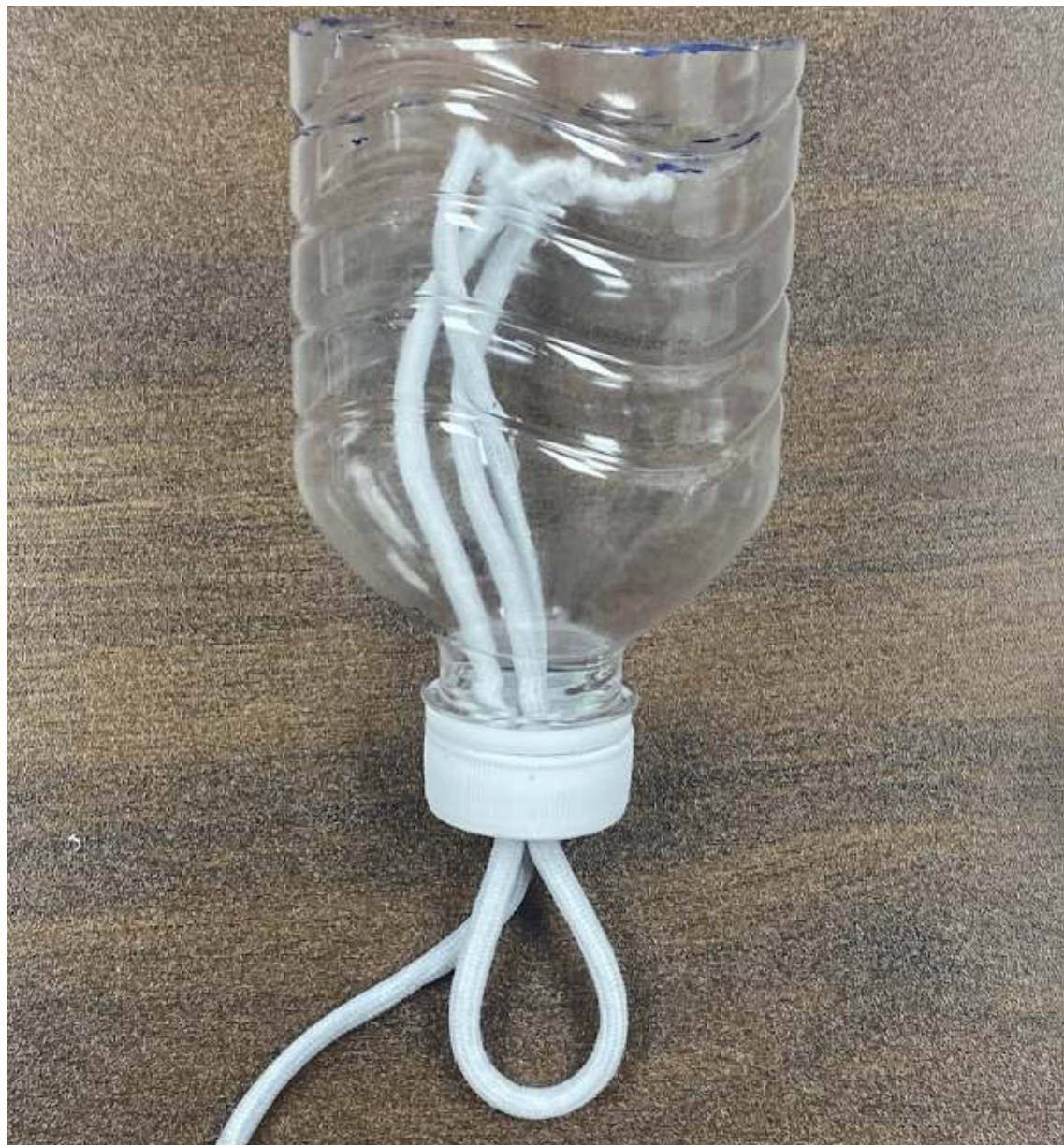
STEP 4

- Take your string and pass it through the hole on the bottlecap
拿起繩子，將它穿過瓶蓋上的洞
- Let the string hang out of both ends of the cap
讓繩子的兩端都從瓶蓋的兩側露出來



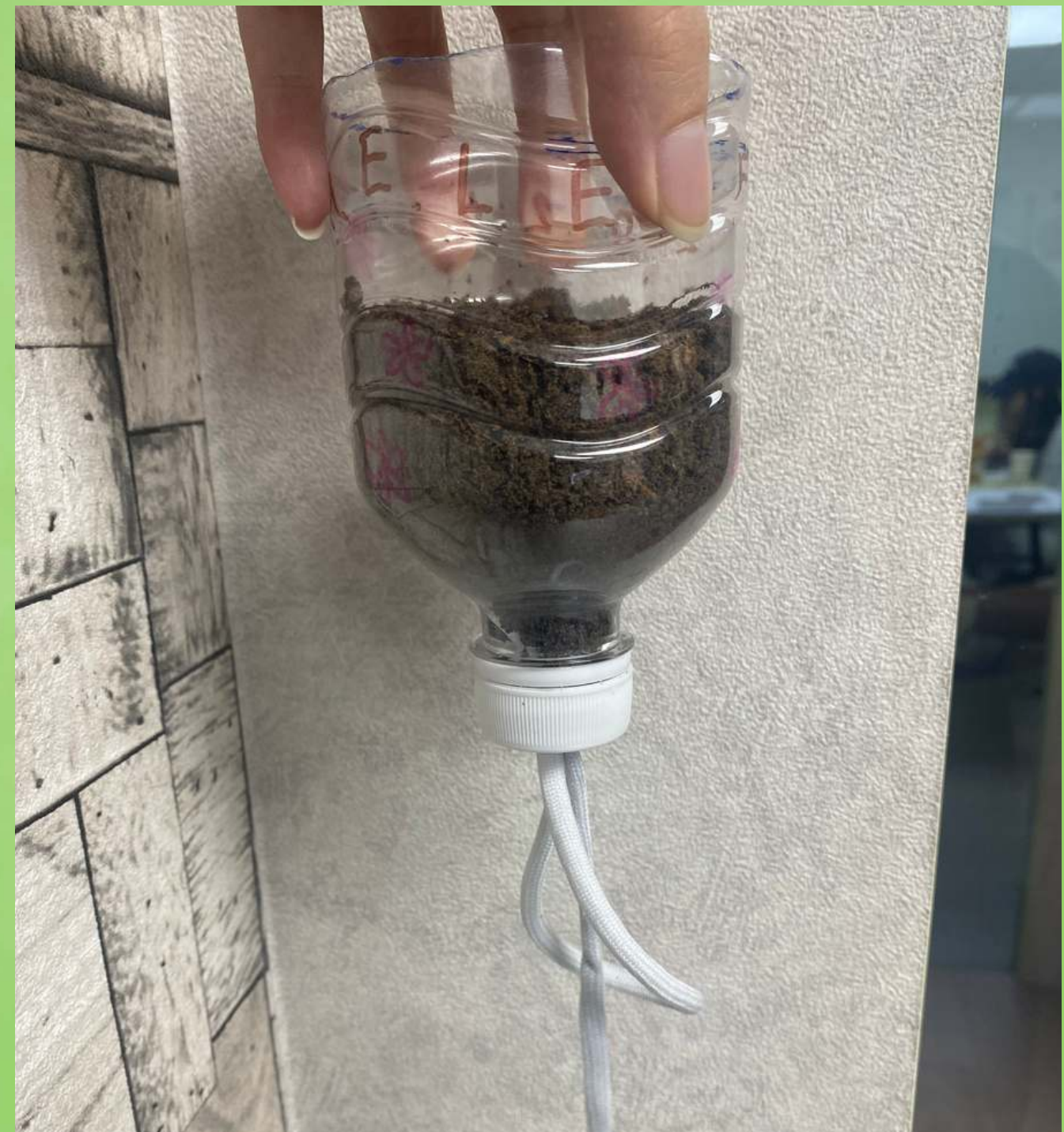
STEP 3

- Put the bottle cap back on the PET bottle and fasten it
把瓶蓋重新蓋回PET寶特瓶並旋緊
- Take the top half of the bottle you cut open and turn it over and put it over the bottom half
將剛剛剪下的上半部分倒過來, 蓋在瓶子的下半部分上



STEP 4

- Take some soil and add it into the top-half of the bottle
拿一些土壤, 加入瓶子的上半部分。



STEP 5

- Take the seed give to you and gently place it into the soil
將發給你的種子輕輕放入土壤中
- Then take more soil and gently cover the seed
然後再取一些土壤，輕輕覆蓋住種子



STEP 6

- In the empty half of the bottle add some water into allow the string to soak up water
在PET寶特瓶的下半部分（空的部分）加入一些水，讓繩子可以吸收水分。

問題：為什麼繩子可以吸收水分？

Summary 課程總結

- Photosynthesis: Plants use sunlight to make their own food to grow.
光合作用：植物利用陽光來製造自己的食物以生長。
- Plant stress: Plants struggle when the environment is too harsh
植物逆境：當天氣太乾、太鹹或太熱時，植物會很難生長。
- DNA: An A-T-C-G code that carries the instructions for all living things.
DNA：由 A-T-C-G 組成的密碼，攜帶著所有生物的生命指令。
- Synthetic Biology: Scientists put new DNA into
living things to make them do new jobs.
合成生物學：科學家將新的密碼放入生物體中，讓它們執行新的任務。





Thank you!

Any Questions ? Feel free to ask us!

有任何問題嗎？歡迎隨時詢問我們！

